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Functool 4 Installation/Service Manual

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DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation *Damage in shipment* written on **all** copies of the freight or express bill **before** delivery is accepted or signed for by a GE Healthcare representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier immediately upon discovery or within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14-day period.

To file a report:

- Call 1-800-548-3366, and use option 8.
- Fill out a report at <http://egems.med.ge.com/edq/home.jsp>.

Contact your local service coordinator for more information on this process.

Rev. 09/17/2006

OMMISSIONS & ERRORS

Customers, contact your GE Healthcare Sales or Service representatives. GE personnel, use the GEMS iTrak Process to report all omissions, errors, and defects in this publication.

IMPORTANT PRECAUTIONS

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient from electric shock, or from mechanical or other hazards.

ПРЕДУПРЕЖДЕНИЕ

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- TENTO PROVOZNÍ NÁVOD EXISTUJE POUZE V ANGLICKÉM JAZYCE.
- V PŘÍPADĚ, ŽE EXTERNÍ SLUŽBA ZÁKAZNÍKŮM POTŘEBUJE NÁVOD V JINÉM JAZYCE, JE ZAJIŠTĚNÍ PŘEKladu DO ODPOVÍDAJÍCÍHO JAZYKA ÚKOLEM ZÁKAZNÍKA.
- NESNAŽTE SE O ÚDRŽBU TOHOTO ZAŘÍZENÍ, ANIŽ BYSTE SI PŘEČETLI TENTO PROVOZNÍ NÁVOD A POCHOPILI JEHO OBSAH.
- V PŘÍPADĚ NEDODRŽOVÁNÍ TÉTO VÝSTRAHY MŮŽE DOJÍT K PORANĚNÍ PRACOVNÍKA PRODEJNÍHO SERVISU, OBSLUŽNÉHO PERSONÁLU NEBO PACIENTŮ VLIVEM ELEKTRICKÉHO PROUDU, RESPEKTIVE VLIVEM MECHANICKÝCH ČI JINÝCH RIZIK.

ADVARSEL

(DA)

- DENNE SERVICE MANUAL FINDES KUN PÅ ENGELSK.
- HVIS EN KUNDES TEKNIKER HAR BRUG FOR ET ANDET SPROG END ENGELSK, ER DET KUNDENS ANSVAR AT SØRGE FOR OVERSÆTTELSE.
- FORSØG IKKE AT SERVICE UDSTYRET MEDMINDRE DENNE SERVICE MANUAL HAR VÆRET KONSULTERET OG ER FORSTÅET.
- MANGLENDE OVERHOLDELSE AF DENNE ADVARSEL KAN MEDFØRE SKADE PÅ GRUND AF ELEKTRISK, MEKANISK ELLER ANDEN FARE FOR TEKNIKEREN, OPERATØREN ELLER PATIENTEN.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εαν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu fáanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다.
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오.
- 이 경고에 유의하지 않으면 전기 쇼크, 기계상의 혹은 다른 위험으로부터 서비스 제공자, 운영자 혹은 환자에게 위해를 가할 수 있습니다.

BRĪDINĀJUMS

(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai angļu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis angļu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente consertar o equipamento sem ter consultado e compreendido este manual de assistência técnica
- O não cumprimento deste aviso pode pôr em perigo a segurança do técnico, do operador ou do paciente devido a choques elétricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kılavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

REVISION HISTORY

Rev	Date	Primary Reasons for Change
1	09/20/05	Initial release.
2	11/21/05	Add FRU Part Number to Section 7
3	06/09/06	Add Installation finish tasks as Section 6 and Update FRU Part Number in Section 8
4	01/15/07	Updated FRU part number in Section 8.
5	06/14/07	Added Korean warning above, changed "cd/" to "cd /" in Section 2-2-2; updated FRU part number in Section 8.
6	02/29/08	Added Omissions & Errors. Updated the following: • Damage in Transportation • Important Precautions Changed the part number in Section 8, FRU List.

TABLE OF CONTENTS

DAMAGE IN TRANSPORTATION	2
OMMISSIONS & ERRORS.....	2
IMPORTANT PRECAUTIONS	3
1- INTRODUCTION	11
1-1 Functool 4 Advanced Application “Clinical” Summary	11
2- PRELIMINARY REQUIREMENTS	11
2-1 License Key Number	11
2-2 Save Configuration	12
2-2-1 Reading Config Media → PC.....	12
2-2-2 Reading Config Media → AW	12
3- INSTALLATION PROCEDURE	12
4- AW AND FUNCTOOL 4 ERRORS.....	13
5- ADVANCED SOFTWARE APPLICATION OPERATION AND DEMO EXAM INFORMATION	13
6- INSTALLATION FINISH TASKS	13
7- SPECIAL CONSIDERATIONS.....	14
8- FRU LIST.....	14

1- INTRODUCTION

The following manual gives a brief overview of the installation process for the Functool 4 advanced application software option. The Functool 4 application is designed to run on the AW4.1 and forward AW platforms only. This guide makes references to all the appropriate standard Advantage Workstation service manuals for more detailed information about installing an advanced application software option on the AW. The current revision of the appropriate AW Service Manual should be used when installing this software. Unique installation processes for Functool 4 and NOT found in the AW service manual can be found in this document in the section, SPECIAL CONSIDERATIONS.

1-1 Functool 4 Advanced Application “Clinical” Summary

Functool 4 is an application that is used to analyze Magnetic Resonance Imaging (MRI) data sets that contain multiple images for each scan location, representing changes in intensity over time (“time series” or “dynamic” data sets), b-value (diffusion imaging), gradient orientation (diffusion tensor imaging) or frequency (spectroscopy).

Functool 4 provides the user with two main tools for analysis:

- Graphs - These show the changes in pixel value over time, b-value or frequency at a given point on the images, either at the cursor position or within a region of interest (ROI) defined by the user.
- Functional Maps - The pixel values from the data set are used to compute a characteristic parameter for every pixel location by means of a function. This can be a basic algorithm or a function derived from such an algorithm. A functional map is then constructed by displaying the value of this parameter for every pixel location.

Functool 4 is supplied in the form of protocols. Each consists of one or more protocol panels that contain the necessary information and the controls required to set up an analysis.

2- PRELIMINARY REQUIREMENTS

2-1 License Key Number

The Functool 4 option requires a license key # based upon the workstation’s Host ID. This key number may be provided by:

- Paperwork shipped with the option that will include one or more 8 to 10-digit license key number(s), created based on the workstation’s/application’s FDO and the workstation’s Host ID/License ID.
- The software package may include a portable media (i.e. floppy disk) with the license key information on it.

The license key information will also be available online at the eLicense website link: <http://elicense.gehealthcare.com/> by searching under the order number or the workstation’s Host ID/License ID.

All of these elements may exist (hard copy license #(s), media, and eLicense data), and at least one of them is necessary to load the software.

2-2 Save Configuration

Create an AW 'CONFIG' media before and after the application software installation for use as needed throughout this and other AW service activities.

2-2-1 Reading Config Media → PC

Configuration information can be read on a PC by viewing the *config.txt* file on the media using text readers like WordPad, Internet Explorer or MS Word. (MS NotePad may not work.)

2-2-2 Reading Config Media → AW

To read a Save Configuration using a floppy disk insert the floppy and type the following in a C shell:

- **su**
- Password: **operator**
- **mount /mnt/floppy**
- **cd /mnt/floppy**
- **more config.txt**
- **cd /**
- **umount /mnt/floppy**

To read a Save Configuration using a CDROM disk insert the CD, and type the following in a C shell:

- **su**
- Password: **operator**
- **mount /mnt/cdrom**
- **cd /mnt/cdrom**
- **more <backup directory name>**
- **cd /**
- **umount /mnt/cdrom**

3- INSTALLATION PROCEDURE

The Standard AW Software Options Installation – Easy Install process applies. For ongoing process revision viability, the details will not be included here, but can be found in the AW Service Manual.

The current revisions for this manual can be viewed online at: <http://aw-ib.euro.med.ge.com>.

1. Open the appropriate Advantage Workstation (AW) Service Manual.
2. Proceed to Job Card titled, *Software Options Installation*.
3. Follow the procedure using the Easy Install method.
4. When complete, perform a *save configuration*.

4- AW AND FUNCTOOL 4 ERRORS

Functool 4 and AW System errors can be examined and investigated in either the usual AW error logs – such as `/var/adm/messages`, and/or `~sdc/logfiles/EasyInstallLog`. Errors specific to Functool 4 can be found in `/export/home/sdc/logfiles/functool.log`.

5- ADVANCED SOFTWARE APPLICATION OPERATION AND DEMO EXAM INFORMATION

To test the functionality of the software, load the DEMO exam CD shipped with the product. The 'Standard' AW Software Demo Exam Installation process applies. For ongoing process revision viability, the details will not be printed here, but can be found in the AW Service Manual.

The current revisions for this manual can be viewed online at: <http://aw-ib.euro.med.ge.com>.

1. Open the appropriate Advantage Workstation (AW) Service Manual.
2. Proceed to Job Card titled, *Advantage Workstation Demo Exam Reinstallation*.
3. Follow the procedure to load the Demo Exams from the CD provided.

To test this application after it is installed, log out and log back into the AW (as suggested by the messages after the installation). Select any data set and click on **Functool** in the *Applications* menu. The *Functool* window will open (on the left side of the screen). This ensures that Functool is installed correctly. Click on the **[Quit]** button to exit Functool.

6- INSTALLATION FINISH TASKS

GIB Update - The Functool4 application must be entered into Global Installed Base (GIB) database after the initial successful installation in order to fulfill product-tracking requirements.

Failure to enter the system and all software applications into GIB will result in GE Healthcare being in violation of FDA tracking compliance, and this system will not be in the affectivity list for future upgrades.

If not familiar with how to get the GIB updated, contact the OLC AW Support. Here is the global GIB link: http://egems.gemedicalsystems.com/gib/gib_entry.jsp

7- SPECIAL CONSIDERATIONS

In addition to Functool, Functool CSI, Functool DTI and Functool FiberTrak will show up at the package installation time as a separate license key entry.

If Functool CSI, Functool DTI, and/or Functool FiberTrak was purchased and they have license keys, they must be selected and the license key numbers entered at the installation prompt for it. Otherwise, leave it unselected.

There can be up to four license keys with this option:

- o Functool **MUST** exist
- o Functool CSI may exist
- o Functool DTI may exist
- o Functool FiberTrak may exist

8- FRU LIST

The replacement part for this option are:

Part #5160880-5 – Functool4 Software and Documentation CD